

# Weather

Current conditions, 48-hour hourly, 5-day, animated radar. No ads, no trackers, no analytics, no accounts, no API keys. Two builds — a web PWA and a native Android app — same data, same look. Repo: [postmaster87/matts-weather-app](https://github.com/postmaster87/matts-weather-app)

Accuracy is ballpark by design — a “do I need a jacket / is that cell going to hit Veenker” app, not a life-safety product. Watches and warnings come straight from the National Weather Service and are the one thing here worth trusting.

## The two builds

|            | Web PWA                                         | Native Android                    |
|------------|-------------------------------------------------|-----------------------------------|
| Where      | repo root, served by GitHub Pages               | android/                          |
| Built with | one HTML + one CSS + one JS file, no build step | Kotlin, Jetpack Compose, osmdroid |
| Install    | Chrome -> Add to Home screen                    | sideload weather.apk              |
| Offline    | service worker + localStorage                   | SharedPreferences cache           |

## What's on the screen

| Section | What it shows                                                                                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alerts  | Active NWS watches/warnings for the exact point. Tap one to read the full text. Hidden when there are none.                                                                                                     |
| Current | Temp, condition, feels-like, today's high/low, wind + direction, gust, humidity, UV, sunrise/sunset.                                                                                                            |
| Hourly  | Next 48 hours, horizontal scroll. The blue bar under each hour is chance of precip.                                                                                                                             |
| 5 Day   | Day, icon, max chance of precip, and a low-to-high bar scaled across the whole 5 days. <b>Tap a day</b> to swap the hourly strip to that day; tap again for the rolling 48 h.                                   |
| Radar   | 10 past frames (~1 hr) plus RainViewer's 30-minute nowcast. Auto-plays. Drag the slider to scrub, <b>Expand</b> for fullscreen, pinch to zoom. A + on the timestamp means a forecast frame, not an observation. |

Location: tap the place name to search any city, or the crosshair/pin to use GPS. The last six places stay as chips in the search screen. Fresh install defaults to Ames, IA.

## Install the Android app

### On your phone, in Chrome:

1. Open <https://postmaster87.github.io/matts-weather-app/weather.apk>
2. Chrome warns about the file type — accept the download.
3. Tap the downloaded file. Android asks to allow installs from Chrome; turn it on, then tap Install.
4. The launcher icon is the sun-behind-cloud, labelled **Weather**.

It is signed with the local debug keystore, so Android calls it an app “from an unknown source.” That is expected for a sideload. Updates must be signed with the same key — install over the top and data is kept; if the key ever changes, uninstall first.

## Install the web app instead

### On your phone, in Chrome:

1. Open <https://postmaster87.github.io/matts-weather-app/>
2. Menu (three-dot) -> **Add to Home screen**

Both keep the last forecast on the device and open offline with an orange **Offline — cached 2:14pm** line at the bottom instead of pretending the numbers are current.

## Rebuilding the Android app

**On the Windows box** — JDK 17, Android SDK 35, build-tools and Gradle 8.9 are already installed there:

```
cd android && ./gradlew assembleRelease
```

Output is android/app/build/outputs/apk/release/app-release.apk (~6.8 MB). Copy it to the repo root as weather.apk to publish it through Pages. android/local.properties holds the SDK path and is not committed.

## Data sources

| Source                                             | Used for                  | Key |
|----------------------------------------------------|---------------------------|-----|
| <a href="#">Open-Meteo</a>                         | Forecast (hourly + daily) | No  |
| <a href="#">Open-Meteo Geocoding</a>               | Place search              | No  |
| <a href="#">api.weather.gov</a>                    | NWS alerts (US only)      | No  |
| <a href="#">RainViewer</a>                         | Radar tiles + nowcast     | No  |
| <a href="#">Esri dark canvas</a>                   | Radar basemap             | No  |
| <a href="#">Leaflet</a> / <a href="#">osmdroid</a> | Map (web / Android)       | No  |

Nothing leaves the phone except the coordinates of the place being looked up. No server, no database, no analytics.

The Android build sends `MattsWeather/1.0 (Android; +<repo URL>)` as its User-Agent, because `api.weather.gov` asks callers to identify themselves and throttles generic agents. It deliberately carries **no**

**contact email** — add one there if you ever want NWS to be able to reach you about traffic.

## Known limits

- **Radar detail caps at zoom 7.** RainViewer's free tiles do not exist past z7 — deeper zooms return a "Zoom Level Not Supported" placeholder. Both builds pin the radar layer at z7 and let the map upscale, so zooming past regional view gets blocky rather than blank. Map zoom is capped at 10.
- **Alerts are US-only.** `api.weather.gov` returns nothing outside the US; the alert strip stays hidden.
- **Radar timestamps are in the phone's timezone**, everything else is in the forecast location's local time. Only matters when looking at another time zone.
- **Daily icon vs. daily precip %** can look inconsistent (an overcast icon next to 89%). Both come from Open-Meteo as-is: the icon is the day's dominant condition, the % is the single wettest hour. Not smoothed over.
- Open-Meteo is a model blend, not a nowcast. Treat the hourly numbers as directionally right, not exact.
- **The Android app is not on Google Play** and has no update mechanism — a new version means building a new APK and installing it over the old one.

## Android source layout

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| <code>MainActivity.kt</code>     | activity, permission request, theme        |
| <code>WeatherVm.kt</code>        | state, refresh + retry, GPS, search        |
| <code>data/Models.kt</code>      | Place, Current, Hour, Day, Forecast, Alert |
| <code>data/WeatherApi.kt</code>  | Open-Meteo, NWS, RainViewer, geocoding     |
| <code>data/Net.kt</code>         | URLConnection with a real User-Agent       |
| <code>data/Store.kt</code>       | SharedPreferences: place, recents, cache   |
| <code>data/Loc.kt</code>         | LocationManager fix + platform geocoder    |
| <code>data/Fmt.kt</code>         | wall-clock string formatting               |
| <code>data/Wmo.kt</code>         | WMO code -> icon + label                   |
| <code>ui/WeatherScreen.kt</code> | screen, top bar, fullscreen radar          |
| <code>ui/Cards.kt</code>         | current / hourly / daily / alert cards     |
| <code>ui/RadarCard.kt</code>     | osmdroid map, frame overlays, timeline     |
| <code>ui/WeatherIcon.kt</code>   | the 12 condition icons, drawn on Canvas    |
| <code>ui/Theme.kt</code>         | palette                                    |

*Icons are drawn on a Compose Canvas using the same 24-unit geometry as the web build's SVGs, so the two versions look identical rather than merely similar.*